

A Privacy-Preserving Provenance Plane for Reconstructable Fast Healthcare Interoperability Resources to Large Language Model Clinical Artificial Intelligence Transactions

Satya Venkata Ranga Janaki Sriram Mentey

Abstract—Clinical artificial intelligence systems that consume electronic health record data and emit reviewer-facing summaries rarely preserve the inputs, transformations, model configuration, evidence, safeguards, and human disposition needed to reconstruct a transaction, while storing raw clinical payloads or hidden model reasoning in an immutable log is unsafe. This paper presents Curie Audit Plane, a protocol and system demonstration for one bounded synthetic Fast Healthcare Interoperability Resources Release 4 to large language model transaction. The plane separates protected clinical content from an append-only audit chain, records typed events with content digests, and checks hash chaining plus an Ed25519 signature over a one-transaction Merkle envelope with a verifier module in the same repository after persisted records are reloaded. Structured model output is constrained to rationale, evidence references, uncertainty, assumptions, and missing data. On synthetic fixtures, reload-and-verify audit reconstruction completeness is 20/20 required fields, and all 19 labeled mutations produce non-VERIFIED outcomes (18 TAMPERED and one INCOMPLETE); access-control probes pass 11/11 allowed or denied outcomes. Capture overhead is reported against an identical unrecorded clinical workflow, not a thin log shortcut. Significance for biomedical research: the method shows how a FHIR-to-LLM handoff can be reconstructed without placing hidden chain-of-thought or raw protected health information in an immutable store. The contribution is an integration of known integrity and interoperability primitives, not a new cryptographic construction.

Index Terms—Biomedical informatics, Fast Healthcare Interoperability Resources, clinical artificial intelligence, provenance, auditability, integrity, health information systems.

I. INTRODUCTION

Large language models are being attached to electronic health record workflows for summarization and review support. Healthcare organizations still need a narrower answer: given one completed transaction, can an investigator reconstruct what the model was shown, how it was configured, which evidence and safeguards were applied, and what a human reviewer decided, without placing raw protected health information or hidden reasoning into an immutable store?

This paper treats that question as a biomedical and health-informatics provenance problem, not as generic software logging. The prototype is deliberately narrow: one synthetic Fast Healthcare Interoperability Resources (FHIR) Release 4

encounter, one structured clinical-summary task, and a human ACCEPT/MODIFY/REJECT disposition. The manuscript is a protocol and system demonstration on synthetic fixtures. It is not a population study, a clinical efficacy trial, or a claim that the implementation is ready for regulatory use.

The frozen research questions are: (i) reconstructability of required provenance fields; (ii) detection of mutation, omission, reordering, broken references, and invalid proofs; (iii) latency and storage overhead of complete capture versus an identical unrecorded clinical workflow; (iv) console exposure of evidence, uncertainty, guardrail results, and disposition as a system demonstration (SCRIPTED_PROXY), not a blinded reviewer study; and (v) replay fidelity for a deterministic stub versus a hosted nondeterministic model.

The intended contribution is that integration: a data/proof split, typed events, structured rationale without hidden chain-of-thought, FHIR Provenance/AuditEvent projections, and reload-and-verify scoring against construct-dependent analytical baselines. Hash chaining and Ed25519 signing are used as known primitives, not as new cryptographic results [1]–[3]. The measured path places one completed transaction root in each Merkle envelope; it does not claim multi-transaction batching.

II. RELATED WORK

Healthcare audit and provenance already have native interoperability surfaces. FHIR Provenance and AuditEvent record agents, entities, and access events, and W3C PROV-DM supplies a generic provenance data model [4]–[6]. IHE Audit Trail and Node Authentication addresses node-level audit trails and secure communication in clinical networks [7]. Those artifacts are necessary interoperability projections. They are not, by themselves, an internal event chain that can reconstruct a FHIR-to-LLM handoff: input manifests, transformations, prompt and model identity, retrieved evidence digests, guardrail outcomes, and human disposition.

Tamper-evident logging has a long cryptographic literature. Haber and Stornetta showed how to time-stamp documents with hash-linked records; Merkle trees supply inclusion proofs; Schneier and Kelsey and Crosby and Wallace developed hash-chained and efficient tamper-evident logs; Certificate Transparency uses Merkle trees for publicly auditable certificate logs [1], [2], [8]–[10]. Ed25519 is a standardized

S. V. R. J. S. Mentey is the corresponding author. Address: 8601 Anderson Mill Rd, Apt 722, Austin, TX 78729, USA (e-mail: srirammentey@ieee.org; ORCID: 0009-0007-2681-006X).

Edwards-curve signature scheme [3]. This prototype composes those primitives in a local append-only store. It is not a distributed ledger and is not evaluated against blockchain-in-healthcare architectures surveyed elsewhere [11].

Clinical large language model evaluation has focused mainly on accuracy, fairness, and the limits of electronic-health-record foundation models [12], [13]. Adapted models can outperform experts on clinical text summarization, and large language models encode substantial clinical knowledge, but those results are not transaction auditability [14], [15]. A systematic review of 519 health-care LLM studies found that almost all used accuracy as the primary dimension and that deployment considerations were rarely measured [13]. That literature does not answer whether a completed FHIR-to-LLM handoff can be reconstructed and integrity-checked without placing raw payloads or hidden reasoning in an immutable log.

Clinical AI governance work addresses transparency to end users and regulators: model cards, clinician-facing model-facts labels, and good machine learning practice for medical devices [16]–[18]. Those artifacts describe intended use and performance. They do not record the inputs, transformations, evidence, guardrails, and human disposition of one completed transaction.

FHIR implementation surveys and the FHIR security framework document how `Provenance`, `AuditEvent`, and related controls are used in interoperable systems [7], [19], [20]. Ordinary application logs, model-provider traces, and EHR access trails typically record who accessed what, or they retain payloads that should not be immutable.

This work is closest to tamper-evident logging combined with a clinical-data boundary. The evaluation comparisons are application JSON logging, hash-only records without signatures, and FHIR-only projection. Those recorders are analytical emulations built for this study; they are not independently shipped products, and their scores are defined against this paper's required field inventory. The prototype does not replace deterministic alert governance, FHIR access control, or the large language model provider.

III. METHOD

A. System Boundary

Curie Audit Plane observes a synthetic FHIR-to-large-language-model workflow. Protected payloads (assembled context, model request and response, reviewer-modified text, and free-form comments) are stored in an authorized content store. The immutable audit store holds metadata, opaque references, schemas, and digests. Reviewer comments on the chain are limited to a controlled category; detailed text is protected. Provider exception text is not stored or fingerprinted. Patient and resource identifiers in immutable metadata are opaque tokens; a protected identity map is required for authorized reconstruction.

B. Event Model and Integrity

A successful transaction emits typed events from input manifestation through human action and an integrity-proof

commitment. Events are canonically hashed and chained. The completed transaction root is represented as a one-leaf Merkle envelope and signed with Ed25519; the measured pipeline does not aggregate multiple transaction roots into one batch. Verification reports `VERIFIED`, `INCOMPLETE`, `TAMPERED`, or `FAILED`. The verifier is a separate module in this repository. Headline reconstructability is computed after reloading persisted SQLite and protected-content records and running that module; it is not an external third-party audit. FHIR Release 4 `Provenance` and `AuditEvent` resources are documented projections of the internal model, not a claimed Implementation Guide profile.

C. Structured Output and Guardrails

Model output must validate as structured rationale. Guardrails record schema validity, evidence-reference coverage, uncertainty presence, a synthetic identifier scan, and policy action. Blocking results require an explicit override policy before `ACCEPT`. Hidden chain-of-thought fields are rejected.

D. Evaluation Protocol

The reproducible command is `curie-audit-plane evaluate`. Reports use schema `curie-evaluation.v1.1` and include experiment metadata (git commit, dirty-tree flag, fixture alias and digest, provider, model, prompt version, decoding parameters, endpoint class, seed, and command template) without absolute local paths as publication identifiers.

Headline audit reconstruction completeness (`independently_verified_arc`) is required-field presence after reloading persisted records, counted only when the in-repository verifier succeeds. The identifier retains the historical metric name; in prose this paper calls the quantity reload-and-verify completeness. Field-presence completeness on the in-memory object is reported separately. Mutation detection uses 19 labeled cases. A case is detected when the verifier returns any non-`VERIFIED` result; the specific status is reported separately. Replay is classified as `EXACT_MATCH`, `EQUIVALENT`, or `DIVERGENT`. Overhead compares the complete plane with an identical unrecorded clinical workflow (load, transform, context, retrieve, complete, guardrails, and an in-memory `ACCEPT` disposition written only to JSONL) on isolated stores with warmup and 30 timed repeats. The unrecorded path does not emit audit events, Merkle proofs, or signatures.

$$\text{overhead}_{\text{latency}} = (T_{\text{plane}} - T_{\text{no_audit}}) / T_{\text{no_audit}}, \quad (1)$$

$$\text{storage_overhead}^{\text{alloc}} = (B_{\text{plane}}^{\text{alloc}} - B_{\text{no_audit}}^{\text{alloc}}) / B_{\text{no_audit}}^{\text{alloc}}, \quad (2)$$

$$\text{storage_total}^{\text{alloc}} = B_{\text{plane}}^{\text{alloc}} / B_{\text{no_audit}}^{\text{alloc}}. \quad (3)$$

Allocated bytes are on-disk file sizes, including SQLite page allocation. Logical bytes are UTF-8 octet lengths of serialized event JSON, transaction-row payloads, and protected-content files, excluding page padding. Relative logical overhead and the logical total multiplier are defined analogously.

Rate metrics use Wilson intervals; the historical report field `latency_ratio` contains the mean paired relative latency overhead. Ablations omit input manifests, transformations, model and prompt metadata, evidence references, cryptographic proofs, or human provenance. Access-control probes record allowed and denied HTTP outcomes for reviewer, investigator, admin, output, content, export, missing transaction, and global list scope. Application JSONL and hash-only JSONL are separately instrumented from the unrecorded workflow; the FHIR projection is built from the source bundle. These recorders are not independently shipped products. The complete plane is scored from the signed audit chain after reload.

The default 50-encounter cohort rewrites one synthetic fixture. It is a single-workflow protocol demonstration, not a population sample. Optional Synthea slices may run from approved roots when dumps exist, but they are excluded from measured evidence because generator version, seed, module set, and CLI are not pinned [21].

E. Threat Model

Adversaries may mutate, delete, or reorder events; substitute proofs or keys; present unsupported findings; or attempt to place protected health information in reviewer comments, provider errors, research exports, or endpoint configuration. The prototype assumes synthetic data, local keys, and an approved loopback model endpoint. It does not claim protection against compromised signing keys or a malicious protected-content store operator.

IV. RESULTS

All numbers below are from the stub evaluation campaign in `evaluation-results/` (schema `curie-evaluation.v1.1`; git commit `0dcf801`, `git_dirty: false`). Software-quality checks at the time of this draft were 146 Python tests, 94.59% statement coverage, and a clean Ruff pass; those are not clinical performance.

A. Reconstructability and Tamper Detection

Reload-and-verify audit reconstruction completeness is 20/20 required fields on the clean fixture transaction after the in-repository verifier succeeds. Field-presence completeness on the in-memory object is also 20/20. Required-event completeness is 11/11 for a successful transaction. All 19 labeled mutations produce a non-`VERIFIED` result: 18 are classified `TAMPERED`, while one substituted content reference is classified `INCOMPLETE` because the referenced protected content is absent. The false-tamper rate over three clean runs is 0. On the cloned 50-encounter cohort, reload-and-verify completeness remains 1.00 with Wilson 95% interval $[0.93, 1.00]$ ($n = 50$) after reloading persisted records.

The complete plane reconstructs more required fields and detects the standard model-event mutation that application logs and FHIR projections do not detect (Table I). Hash-only logging detects digest mismatch without providing chain

TABLE I
BASELINE COMPARISON ON THE SYNTHETIC FIXTURE (COMPLETE PLANE VERSUS CONSTRUCT-DEPENDENT UNRECORDED-WORKFLOW AND SOURCE-BUNDLE RECORDERS). SCORES USE THIS PAPER'S REQUIRED-FIELD INVENTORY.

Recorder	ARC	Tamper detection	Queryability
Application JSONL	0.10	0.00	0.00
Hash-only JSONL	0.15	1.00	0.00
FHIR R4 projection	0.20	0.00	0.25
Complete plane	1.00	1.00	1.00

TABLE II
RECONSTRUCTABILITY AFTER OMITTING A PROVENANCE CLASS.

Condition	ARC	Δ from full
Full recorded provenance	1.00	0.00
Omit input manifests	0.95	0.05
Omit transformations	0.95	0.05
Omit model metadata	0.90	0.10
Omit evidence references	0.95	0.05
Omit cryptographic proofs	0.75	0.25
Omit human provenance	0.85	0.15

or signature guarantees. These gaps are expected from the construct: the baselines were not designed to record the same field set.

B. Ablations

Omitting cryptographic proofs drops reconstructability from 1.00 to 0.75 ($\Delta = 0.25$). Omitting human provenance drops it to 0.85 ($\Delta = 0.15$). Omitting model and prompt metadata drops it to 0.90. Omitting input manifests, transformations, or evidence references each drops one required field (0.95). These are field-presence ablations of recorded provenance classes, not clinical-efficacy ablations (Table II).

C. Access Control

Eleven role probes passed with observed HTTP status equal to the predeclared expectation (11/11; Wilson 95% interval $[0.74, 1.00]$): reviewer read and output allowed; reviewer export and investigator output or content denied; investigator verify and export allowed; admin content allowed; unauthenticated list denied (401); missing transaction reported as missing (404); global list allowed.

D. Overhead and Replay

Mean plane latency is 16.13 ms versus 8.59 ms for the identical unrecorded workflow. The mean of the 30 paired relative latency overheads is 1.033 (+103.3%; 95% interval $[+78.8\%, +127.8\%]$ after one warmup). Relative allocated storage overhead is 6.20 (70 512 additional bytes; 81 885 versus 11 373 bytes). Total allocated bytes are $7.20\times$ the baseline. The allocated audit file is one SQLite page (65 536 bytes); logical serialized event and transaction payloads are 23 105 bytes. Logical plane storage is 39 454 versus 11 373 baseline bytes (relative logical overhead 2.47; total logical multiplier $3.47\times$). These values do not support a claim that audit capture is under 15% of an already-complete clinical application.

Verification of the clean transaction is 1.55 ms. Deterministic-stub same-prompt replay is exact; prompt-substitution replay is DIVERGENT by construction. The frozen live campaign records one hosted same-prompt replay observation, classified as DIVERGENT ($n = 1$). This draft does not estimate a hosted-replay rate.

The 16-arm scenario matrix exercises AC-CEPT/MODIFY/REJECT, forced and natural WARN/BLOCK, provider failure, sparse encounter, Synthea slices, MODIFY evidence, prompt-substitution replay, and access audit. It is workflow coverage, not a population sample. The two Synthea slice arms are unpinned demonstrations and are excluded from measured tables. Console reconstruction is SCRIPTED_PROXY only.

V. LIMITATIONS

This is a single-workflow protocol demonstration on synthetic FHIR fixtures. The default cohort rewrites one fixture; it is not 50 independent patients and does not support population-style inference. Baseline ARC and tamper scores are construct-dependent: they measure this paper's required-field inventory against analytical emulations, not independently shipped logging products. Reload-and-verify completeness uses an in-repository verifier, not an external audit. Mutation detection is a non-VERIFIED outcome; 18/19 cases were specifically TAMPERED. Each measured Merkle envelope contains one transaction; batching is unevaluated. SQLite page size confounds allocated storage, so logical bytes are less confounded; multi-transaction amortization is unreported. Latency $n = 30$ is an overhead sketch, not a capacity or tail-latency study. FHIR projections are mapping rules, not Implementation Guide profile validation. Synthea generator version and CLI parameters are NOT_PINNED; optional slice arms are excluded from measured evidence. A DOI for the public archive is not assigned in this draft.

VI. ETHICS, FUNDING, AND DISCLOSURE

No human subjects or identifiable clinical records were used. Evaluation uses synthetic templates and optional external Synthea bundles [21]; therefore no institutional review board protocol was required. The author reports no funding and no conflict of interest. Portions of software and documentation were drafted with assistance from coding agents; evaluation metrics are produced by the versioned Python harness. The author remains responsible for claims, code, and the decision to submit.

VII. CONCLUSION

A privacy-preserving provenance plane can reconstruct a bounded synthetic FHIR-to-large-language-model transaction from persisted records with an in-repository verifier, detect a defined mutation suite, and keep hidden reasoning and free-form reviewer text out of the immutable chain. The result is a protocol demonstration of known primitives under a clinical data/proof split, not a new integrity scheme and not a population evaluation. Remaining work is independent synthetic-patient scale, amortized batching, hosted-model replay characterization, and IEEE Author Portal packaging.

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